



**ALLIANCE
UNIVERSITY**
Private University established in Karnataka State by Act No. 34 of year 2010
Recognized by the University Grants Commission (UGC), New Delhi

**NAAC
GRADE A+**
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

COURSE : Deep Learning

Project : 04

Title: Ethical Considerations in Deep Learning

Submitted By

Name: Tejas Pebam

Reg. No.: 2411021240018

Submitted To:

Faculty Name: Nagasri P.

Department of Computer Application

Alliance University

Chandapura – Anekal Main Road, Anekal

Bengaluru – 562106

Ethical Considerations in Deep Learning

Introduction

Deep Learning now powers facial recognition, medical diagnosis, credit scoring, recommendation systems, and self-driving cars. As these systems increasingly influence decisions that affect people's lives, jobs, and freedom, examining the ethics of how they are designed, trained, and deployed has become essential. Unlike traditional software, Deep Learning learns behavior from data rather than fixed rules, so its outputs are only as trustworthy as the data and deployment practices behind them. This report examines three major ethical issues — privacy, transparency, and responsible use — analyzes a real-world case in which they caused measurable harm, and proposes measures to reduce these risks.

Task 1: Three Ethical Issues in Deep Learning

1. Privacy Concerns

Deep Learning systems are typically trained on massive datasets that include images, voice recordings, browsing history, or biometric data — much of it personal and sensitive. Collecting and using such data without clear consent or adequate safeguards creates serious privacy risk. Training data is often scraped from the internet without meaningful consent, and trained models can sometimes memorize and leak fragments of their training data (model inversion or membership inference), letting an attacker determine whether a specific person's data was used to train the model. Even a model's outputs — a health score, a facial match, a targeted ad — can themselves reveal sensitive information about a person.

2. Lack of Transparency (The Black-Box Problem)

Many Deep Learning models, especially deep neural networks with numerous hidden layers, are called "black boxes" because even the engineers who built them often cannot explain exactly why a model produced a specific output. This makes it difficult to detect when a model relies on inappropriate signals, difficult for affected individuals to contest a decision made about them, and difficult for regulators to hold organizations accountable. Transparency is closely tied to accountability and public trust — without it, problems can stay hidden for years.

3. Responsible Use of AI

Even a well-built model can cause harm if deployed carelessly, used outside its intended scope, or applied without human oversight. Responsible use concerns the decisions made around a model — who deploys it, in what context, and with what safeguards. Irresponsible use includes deploying a model trained on one population onto a very different one without revalidation, treating AI output as a final decision instead of one input to a human, deploying surveillance tools without legal review, and failing to monitor a system after launch for drift or unintended harm.

Task 2: Real-World Example – Clearview AI

A well-documented case illustrating all three issues is Clearview AI, a facial recognition company that built a searchable database by scraping billions of photographs from social media and other public webpages, without the knowledge or consent of the people pictured, and sold access to its search tool to law enforcement agencies and private companies.

Investigations from 2020 onward revealed that Clearview scraped images at massive scale in violation of platform terms of service. Its internal matching logic was never made public, so neither the people affected nor independent auditors could examine how matches were generated or how error-prone the system was. Regulators in Canada, the UK, France, Italy, and Australia later ruled the data collection violated privacy law, imposing fines and ordering deletion of citizens' data; in the US, the company faced biometric-privacy lawsuits and agreed to restrict most commercial sales.

- **Privacy:** Billions of facial images were collected and used without consent, turning public photos into a permanent, searchable biometric identifier.
- **Lack of Transparency:** The matching algorithm and database were kept secret, so no one could verify how the tool worked or how often it was wrong before investigations forced disclosure.
- **Responsible Use:** The tool was sold to police and used to identify individuals — including cases of wrongful identification — without adequate legal oversight or accuracy testing.

Task 3: Measures to Address the Identified Issues

Protecting Privacy

- Practice data minimization and obtain clear, informed consent before collecting personal data.
- Use privacy-preserving techniques: anonymization, differential privacy, and encryption.
- Comply with data protection laws such as the GDPR, and limit data retention periods.

Improving Transparency

- Use Explainable AI (XAI) techniques so developers and affected individuals understand why a decision was made.
- Allow independent, third-party audits of high-stakes systems.
- Publish model documentation ("model cards"/"datasheets") disclosing intended use, limitations, and known error rates.

Ensuring Responsible Use

- Define clear accountability for harm caused by an AI system, with mechanisms to appeal automated decisions.
- Require human review before AI output drives high-stakes decisions.
- Revalidate models before deploying into new contexts or populations, and add legal review before deploying sensitive technologies like surveillance.

Conclusion

As the Clearview AI case shows, the ethical risks of Deep Learning are not theoretical. Privacy violations, lack of transparency, and irresponsible deployment are interconnected problems that can combine to cause serious real-world harm at scale. Addressing them requires privacy-protective engineering, explainable and auditable models, and clear accountability — combining technical safeguards with strong policy and independent oversight to deploy Deep Learning systems more safely and fairly.